Experiment 9:

Use CDH (Cloudera Distribution for Hadoop) and HUE (Hadoop User Interface) to analyze data and generate reports for sample datasets

**1. Upload Student Marks CSV to HDFS**
Sample file: student_marks.csv

```
student_id,name,department,sub1,sbu2,sub3
cd101,John Doe,Engineering,88,92,85
cd102,Jane Smith,Science,78,85,80
cd103,Ravi Kumar,Engineering,90,91,89
cd104,Ayesha Ali,Science,65,70,75
```

**In HUE:**

1. Go to **File Browser**.
2. Create a folder: /student_data/
3. Upload student_marks.csv into that directory.

**2. Create External Hive Table in HUE**

```
CREATE EXTERNAL TABLE IF  student_marks (
  student_id STRING,
  name STRING,
  department STRING,
  sub1 INT,
  sub2 INT,
  sub3 INT)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
LOCATION '/student_data';
```

**3. Run Analysis Queries:**

**Average Marks per Subject**

```
SELECT
  AVG(sub1) AS avg_math,
  AVG(sub2) AS avg_physics,
  AVG(sub3) AS avg_chemistry
FROM student_marks;
```

**Output:**

| avg_math | avg_physics | avg_chemistry |
|---|---|---|
| 80.25 | 84.5 | 82.25 |

**Total and Average Marks per Student**

```
SELECT student_id,name,(sub1+ sub2 + sub3) AS total_marks,
  ROUND((sub1+ sub2 + sub3)/3.0, 2) AS avg_marks
FROM student_marks;
```

**Output:**

| student_id | name | total_marks | avg_marks |
|---|---|---|---|
| 101 | John Doe | 265 | 88.33 |
| 102 | Jane Smith | 243 | 81.00 |
| 103 | Ravi Kumar | 270 | 90.00 |
| 104 | Ayesha Ali | 210 | 70.00 |

**Top 2 Students by Total Marks**

```
SELECT name,(sub1+ sub2 + sub3) AS total FROM student_marks
ORDER BY total DESC LIMIT 2;
```

**Output:**

| name | total |
|---|---|
| Ravi Kumar | 270 |
| John Doe | 265 |

**Department-Wise Average**

```
SELECT
  department,
  ROUND(AVG(sub1),2) AS avg_math,
  ROUND(AVG(sub2),2) AS avg_physics,
  ROUND(AVG(sub3),2) AS avg_chemistry
FROM student_marks
GROUP BY department;
```

**Output:**

| department | avg_math | avg_physics | avg_chemistry |
|---|---|---|---|
| Engineering | 89.00 | 91.50 | 87.00 |
| Science | 71.50 | 77.50 | 77.50 |